

HOSPITAL PATIENT ANALYTICS REPORT

A storytelling analysis of patient volume, treatment cost, length of stay, readmission, outcomes and satisfaction

984	8.23M	8.37K	37.66 days	26.83%	3.60 / 5
Patients	Total treatment cost	Avg cost per patient	Avg length of stay	Readmission rate	Avg satisfaction

Executive narrative

The data tells the story of a hospital managing **984 patient records** across 15 medical conditions and 15 treatment procedures. The overall picture is not one of a single uniform patient journey. Instead, it shows a service environment shaped by high-cost oncology and cardiac care, a largely adult cohort, and a meaningful readmission burden concentrated in a small number of conditions.

Across the cohort, treatment cost totals **8,233,600** in the dataset's currency units. The average cost per patient is **8,367**, while the typical recorded stay averages **37.66 days**. Readmissions affect **264 patients (26.83%)**. Although **591 patients (60.06%)** are recorded as recovered, the data points to specific areas where discharge planning, pathway review and patient-experience monitoring deserve attention.

Prepared for data analytics portfolio documentation

1. Purpose, scope and analytical approach

This report translates the hospital dataset into a decision-focused narrative. It combines three layers of work: a data-quality review, a SQL Server analysis containing more than 20 analytical questions, and an interactive Power BI dashboard with measures, calculated categories and visual drill-downs.

The objective was to understand the scale of the patient population, where treatment spending is concentrated, which conditions are associated with long stays and readmissions, and how outcomes and satisfaction vary across the service mix.

Data quality assessment

Check	Result	Interpretation
Records and fields	984 records; 13 fields	Sufficient coverage for descriptive analysis across patient, cost and outcome variables.
Completeness	No missing values identified	The analysis did not require imputation.
Duplicates	No duplicate rows identified	Each record can be treated as a distinct observation.
Patient IDs	Unique for all records	Patient counts can be calculated with confidence.
Category consistency	Gender, Outcome and Readmission codes were standardized	Groupings and calculations are reliable.

Method note: the source has no date field, so this report evaluates the cohort as a whole rather than showing trends over time. Cost values are reported in the dataset's original currency units; no currency code was supplied. Readmission was stored as a yes/no field in the source data and as a bit field after SQL import (1 = readmitted, 0 = not readmitted).

The dataset describes associations, not causes. For example, a high readmission rate for a condition indicates an operational area to investigate; it does not prove that the condition itself caused readmission.

2. Dashboard overview

The Power BI dashboard brings the SQL findings into one operational view. KPI cards establish scale immediately, while condition-level visuals help the user compare patient volume, cost, readmission risk, satisfaction, and length of stay. Slicers for gender, age group, condition, procedure, readmission and outcome allow the story to be explored without changing the model.

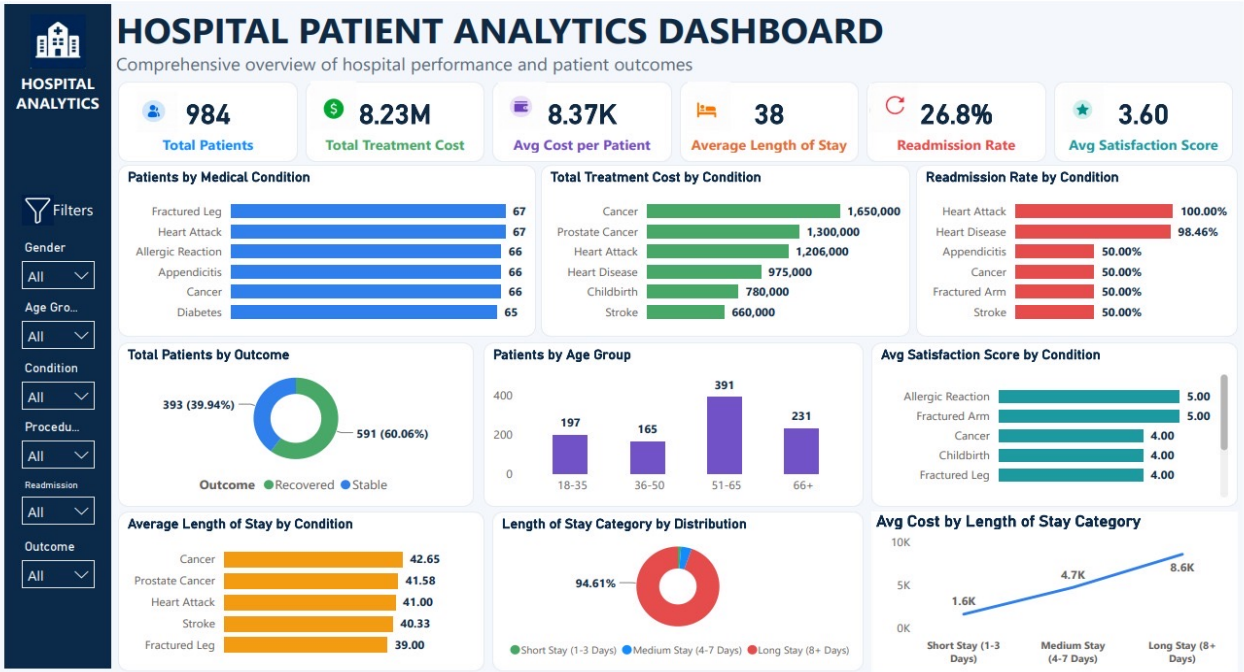


Figure 1. Final Power BI Hospital Patient Analytics Dashboard.

3. Who is being served? The patient population story

The cohort is predominantly middle-aged and older. The average age is **53.75** years (median 55), with ages ranging from 25 to 78. The largest age group is **51-65 years** with **391 patients**, followed by **66+ years** with **231**. There are no patients under 18 in the source data, so the findings should be understood as an adult-cohort analysis.

Patient volume is relatively even across conditions: the most frequent conditions, Fractured Leg and Heart Attack, each have 67 patients. This balance is useful: observed cost or quality differences are not simply the result of one condition dominating the patient base.

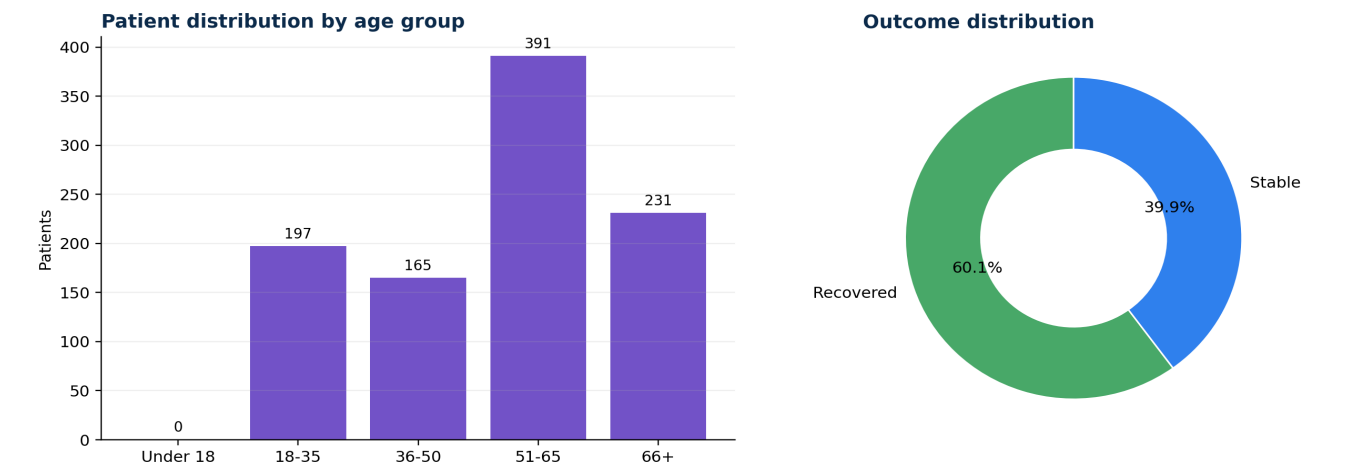


Figure 2. Age mix and outcome distribution.

Age group	Patients	Share
Under 18	0	0.0%
18-35	197	20.0%
36-50	165	16.8%
51-65	391	39.7%
66+	231	23.5%

4. The cost story: a small set of conditions drives most spend

Total treatment cost is **8,233,600**. Spending is concentrated in a small set of higher-acuity conditions. Cancer is the largest cost contributor at **1,650,000**, followed by Prostate Cancer at **1,300,000** and Heart Attack at **1,206,000**. Together, these three conditions account for **50.5%** of all recorded treatment cost.

The procedure view reinforces the same pattern. Surgery and Chemotherapy has the highest average procedure cost at **25,000**, followed by Radiation Therapy at **20,000** and Cardiac Catheterization at **18,000**. These are logical targets for cost-monitoring conversations because even small improvements in pathway efficiency could have a material effect on total expenditure.

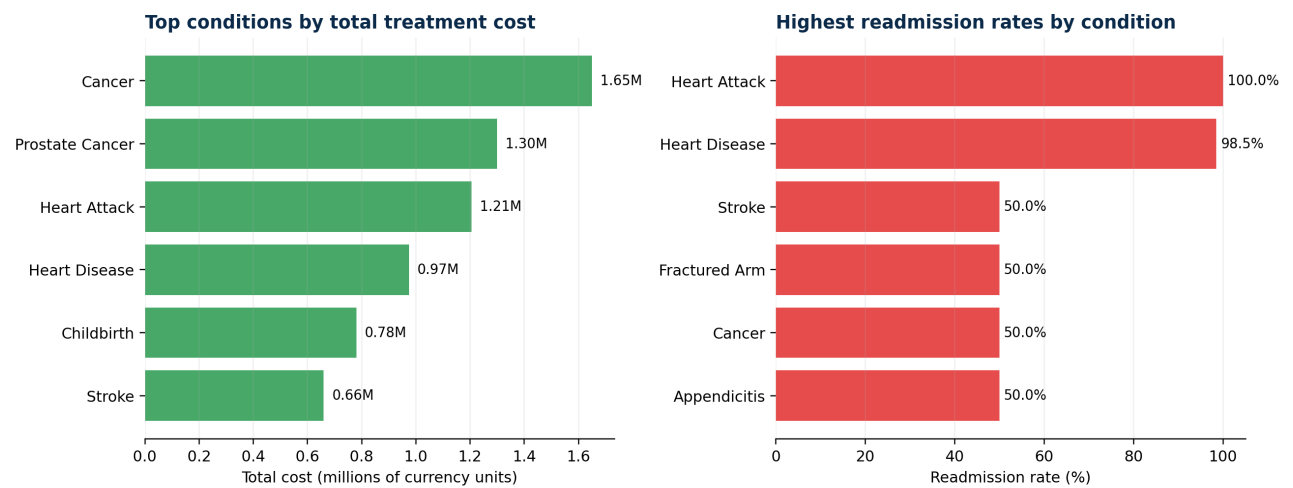


Figure 3. Cost concentration and condition-level readmission risk.

Condition	Patients	Total cost	Avg cost
Cancer	66	1,650,000	25,000
Prostate Cancer	65	1,300,000	20,000
Heart Attack	67	1,206,000	18,000
Heart Disease	65	975,000	15,000
Childbirth	65	780,000	12,000
Stroke	66	660,000	10,000

5. Length of stay: the operational pressure point

Length of stay is a defining feature of this dataset. The average is **37.66 days**, and **931 patients (94.6%)** fall into the long-stay category. This concentration explains why the average stay is high and suggests that bed capacity, discharge coordination and care transitions should be central operational questions.

Cancer has the longest average stay at **42.65 days**, followed by Prostate Cancer at **41.58 days**, Heart Attack at **41.00 days**, and Stroke at **40.33 days**. These pathways are also among the more expensive ones, indicating a combined capacity and financial impact.

The stay-category analysis shows that average cost rises from **1,617** for short stays to **8,614** for long stays. Satisfaction decreases from **4.42** to **3.58**, while readmission increases from **16.7%** to **27.4%**. This does not establish causation, but it gives a practical reason to review the long-stay experience.

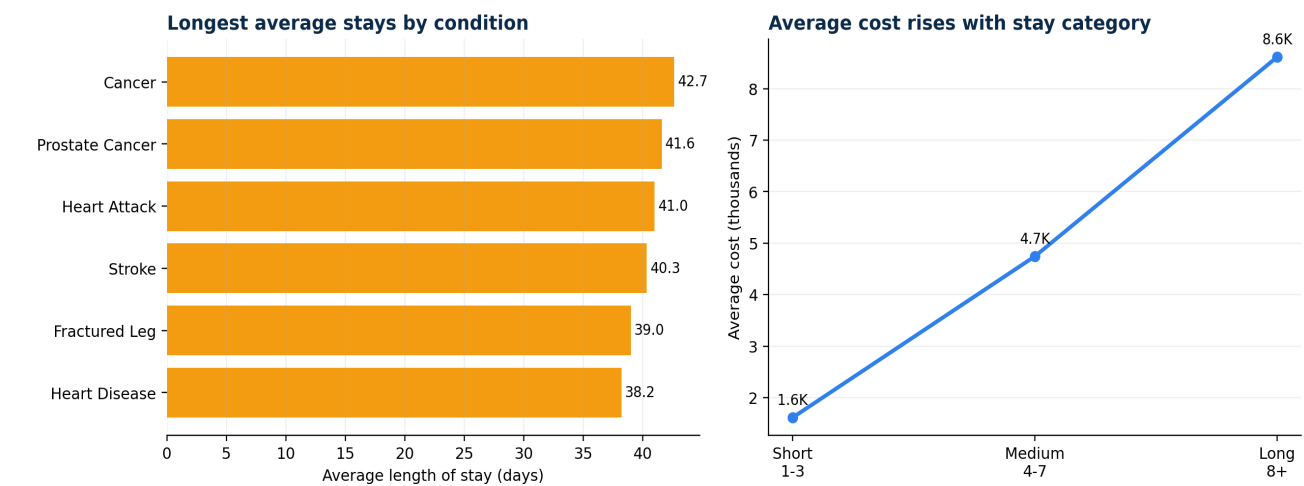


Figure 4. Conditions with the longest average stay and cost by stay category.

6. Readmission: a concentrated risk, not a uniform problem

The overall readmission rate is **26.83%**, representing **264** readmitted patients. The headline rate is important, but the condition-level view is more useful: Heart Attack has a **100.00%** rate (67 of 67 records), and Heart Disease has a **98.46%** rate (64 of 65). Appendicitis, Cancer, Stroke and Fractured Arm each show a 50.00% rate. By contrast, several conditions have no recorded readmissions in this dataset.

The high-cost, long-stay, readmitted query identifies **196** records meeting all three risk flags. These records should be treated as a review cohort rather than an automatic intervention list: they can be used to test whether discharge education, medication follow-up, referral timing or care-coordination gaps are contributing to repeat admissions.

Readmission also differs by gender in this dataset. Female patients have a rate of **43.89%** (230 of 524), compared with **7.39%** among male patients (34 of 460). This finding should be validated with case-mix, age, diagnosis and pathway context before any conclusion about inequality or causation is made.

Condition	Patients	Readmitted	Readmission rate
Heart Attack	67	67	100.00%
Heart Disease	65	64	98.46%
Appendicitis	66	33	50.00%
Fractured Arm	66	33	50.00%
Cancer	66	33	50.00%
Stroke	66	33	50.00%
Diabetes	65	1	1.54%
Allergic Reaction	66	0	0.00%



Figure 5. Satisfaction leaders and gender-level readmission comparison.

7. Outcomes and satisfaction: recovery is strong, experience varies

Overall, **591 patients (60.06%)** are marked as recovered and **393 (39.94%)** as stable. Recovered patients also have a higher average satisfaction score (3.84) than stable patients (3.24). This suggests that the dataset's outcome status and patient experience move together, though further clinical and survey context would be required to explain why.

The overall satisfaction score is **3.60 out of 5**. Allergic Reaction and Fractured Arm each score **5.00**, while Heart Attack and Stroke each average **2.00**. Cardiac Catheterization and CT Scan and Medication, the procedures linked to these lower-satisfaction pathways, also average 2.00. These service lines merit a patient-journey review focused on communication, expectation setting, pain management, follow-up and handover quality.

The key lesson is that a favorable aggregate recovery rate should not hide variation in patient experience. Reviewing high-cost, high-readmission and low-satisfaction pathways together provides a stronger operational lens than reviewing any one measure in isolation.

Condition	Recovery rate	Avg satisfaction	Avg stay	Total cost
Cancer	100.0%	4.00	42.65	1,650,000
Heart Attack	0.0%	2.00	41.00	1,206,000
Heart Disease	100.0%	3.49	38.22	975,000
Stroke	0.0%	2.00	40.33	660,000
Fractured Arm	100.0%	5.00	34.26	33,000
Allergic Reaction	100.0%	5.00	34.24	6,600

8. Recommendations and next actions

The analysis suggests a focused, practical response rather than a broad attempt to change every pathway at once.

1. Establish a cardiac readmission review

Heart Attack and Heart Disease account for the highest observed readmission rates. Review discharge criteria, medication reconciliation, early follow-up appointment completion and escalation pathways for these patients.

2. Create a long-stay management list

More than nine in ten records fall in the long-stay category. Start with Cancer, Prostate Cancer, Heart Attack and Stroke to map the reasons for longer stays and identify controllable delays.

3. Protect high-cost oncology and cardiac pathways

Cancer, Prostate Cancer and Heart Attack represent a large share of treatment cost. Add pathway-level cost monitoring, utilisation review and variance analysis before considering cost-reduction actions.

4. Investigate low-satisfaction high-acuity journeys

Heart Attack, Stroke, Cardiac Catheterization and CT Scan and Medication have low satisfaction. Combine feedback review with patient-journey mapping to identify where experience breaks down.

5. Make the dashboard operational

Use the Power BI filters in routine meetings to review the high-cost, long-stay and readmitted cohort by condition, procedure, gender and outcome. The aim is earlier identification, not retrospective reporting only.

6. Improve future data capture

Add admission/discharge dates, department, clinician/team, severity indicators, discharge destination and survey detail. These fields would allow more reliable trend, root-cause and predictive analysis.

Conclusion

The hospital dataset shows a mixed operational picture: positive recovery outcomes for a majority of patients, but a long-stay-heavy cohort, concentrated high-cost pathways, and sharply elevated readmission risk in specific cardiac conditions. SQL made these patterns measurable; Power BI made them easy to explore. Together, the work provides a practical foundation for performance review, care-pathway prioritisation and future data-driven improvement.

Appendix A - SQL analysis coverage

The SQL Server script was structured around more than 20 analytical questions. The following groups summarize the scope of the analysis.

Analysis area	Examples of questions answered
Population profile	Total patients; average age; gender distribution; most common conditions and procedures.
Cost analysis	Total cost; average cost per patient; total cost by condition; average and total procedure cost.
Length of stay	Overall average stay; longest average stay by condition; stay-category cost and satisfaction comparison.
Readmission and outcomes	Overall readmission; readmission by condition and gender; outcome distribution; recovery rate by condition.
Experience and risk	Satisfaction by condition and procedure; high-cost, long-stay, readmitted patient cohort.

Appendix B - Selected evidence from the SQL analysis

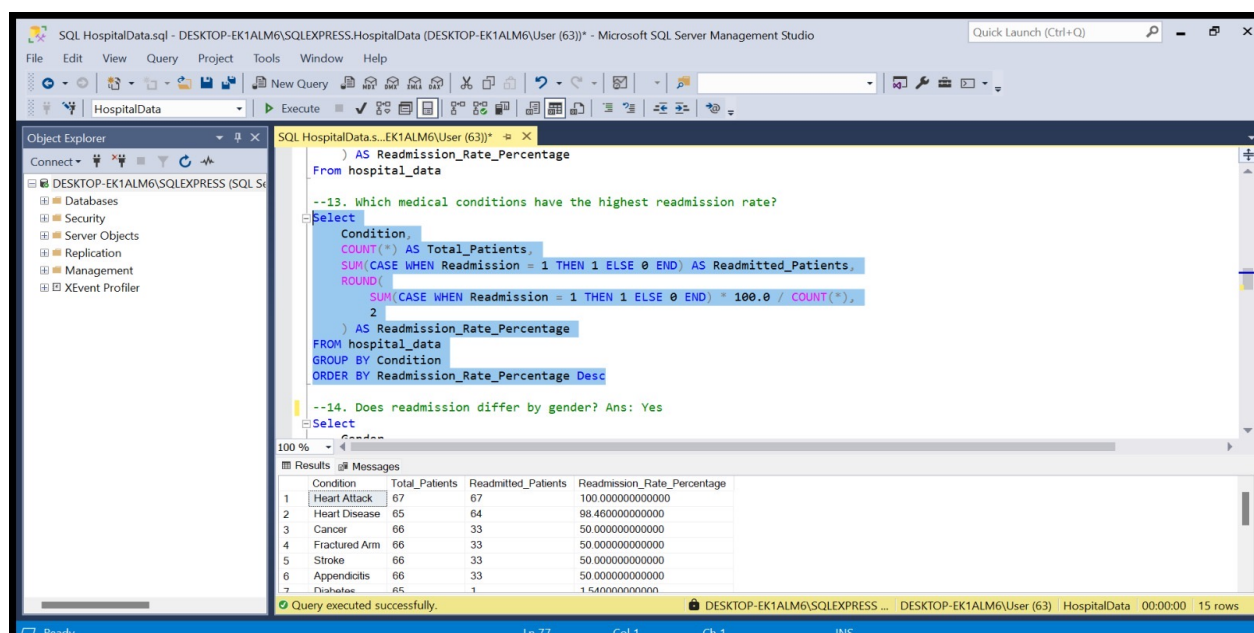


Figure B1. High-cost, long-stay and readmitted patient query.

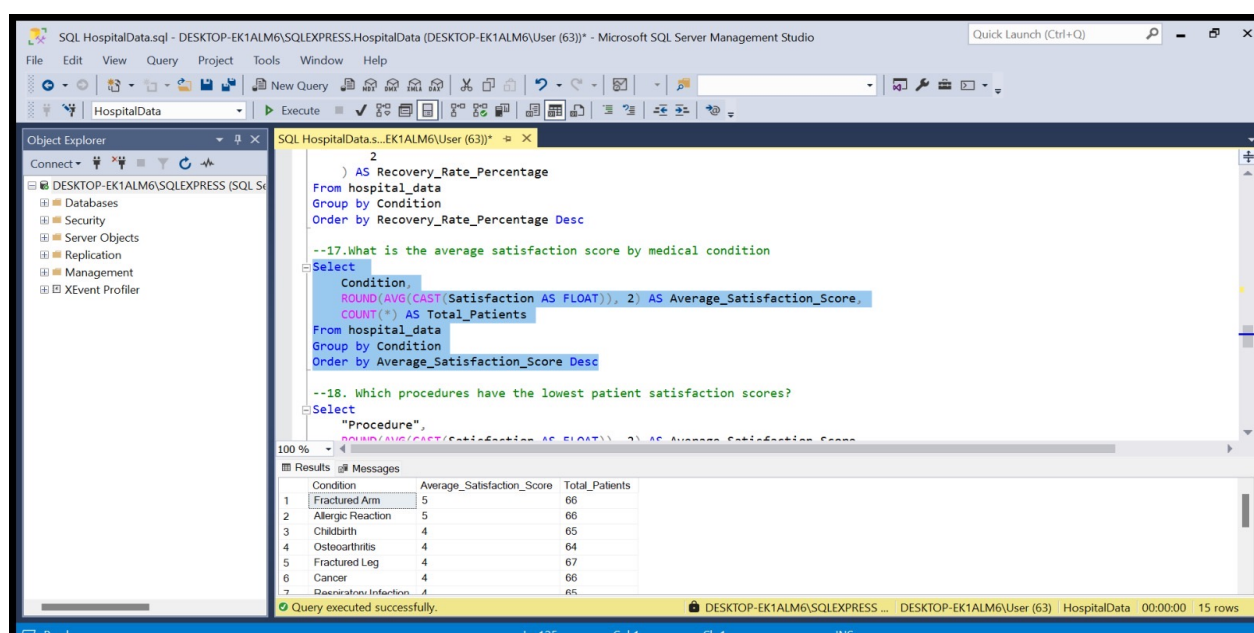


Figure B2. Satisfaction score by medical condition query.

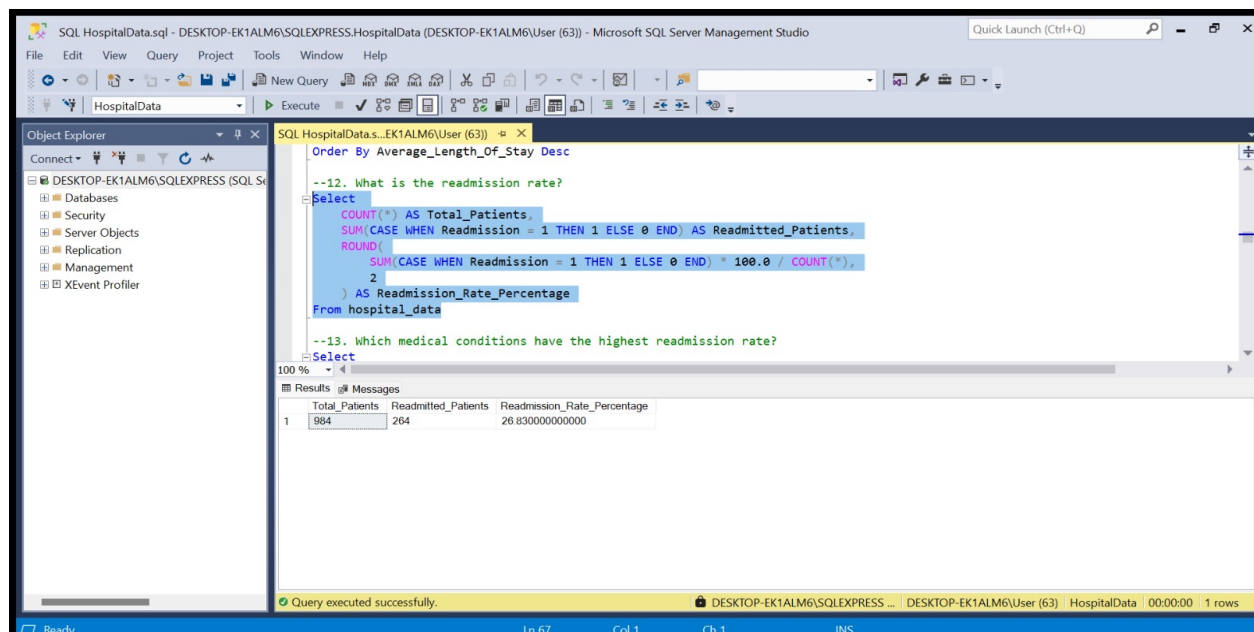


Figure B3. Readmission rate by condition query.